

Raw Matrices A and B:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Step-by-Step Multiplication:

$$C = \begin{bmatrix} (1 \times 5 + 2 \times 7) & (1 \times 6 + 2 \times 8) \\ (3 \times 5 + 4 \times 7) & (3 \times 6 + 4 \times 8) \end{bmatrix}$$

Interpretation with Matrix Indexing:

- $C[0, 0] = A[\text{row } 0] \times B[\text{col } 0] = 1 \times 5 + 2 \times 7 = 19$
- $C[0, 1] = A[\text{row } 0] \times B[\text{col } 1] = 1 \times 6 + 2 \times 8 = 22$
- $C[1, 0] = A[\text{row } 1] \times B[\text{col } 0] = 3 \times 5 + 4 \times 7 = 43$
- $C[1, 1] = A[\text{row } 1] \times B[\text{col } 1] = 3 \times 6 + 4 \times 8 = 50$

Final Matrix C:

$$C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Summation for Each Entry of C:

$$C[i, j] = \sum_{k=0}^{N-1} A[i, k] \times B[k, j]$$

Dot Product Definition:

The dot product of two vectors **a** and **b** of the same size is defined as:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i$$

where a_i and b_i are the components of vectors **a** and **b**, respectively, and n is the number of components in the vectors.

This definition of dot product is what each entry of matrix C represents: the dot product of a row from matrix A and a column from matrix B .